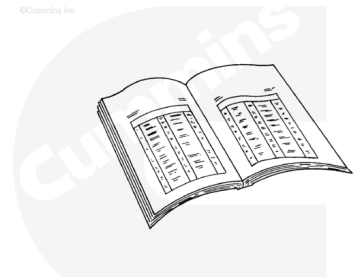


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



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⚠ WARNING ⚠

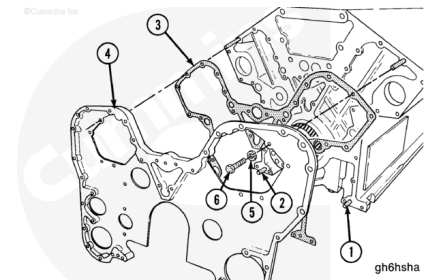
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative battery cable last.

- Disconnect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Remove the front gear cover. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/28/28-001-031-tr.html>)
- Remove the camshaft gears. Refer to Procedure 001-012 in Section 1. (</qs3/pubsys2/xml/en/procedures/28/28-001-012-tr.html>)
- Remove the idler shaft and gears. Refer to Procedure 001-083 in Section 1. (</qs3/pubsys2/xml/en/procedures/28/28-001-083-tr.html>)
- Clean and inspect for reuse idler shafts and camshaft idler gears. Refer to Procedure 001-061 in Section 1. (</qs3/pubsys2/xml/en/procedures/28/28-001-061-tr.html>)
- Clean and inspect for reuse idler shafts and water pump idler gear. Refer to Procedure 001-063 in Section 1. (</qs3/pubsys2/xml/en/procedures/28/28-001-063-tr.html>)

Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove the 15 capscrews (6).

Remove the gear housing (4) and the gasket (3).

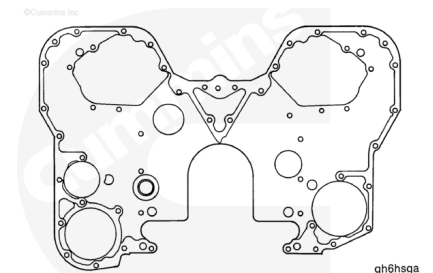
Discard the gasket.

The dowel pins do **not** have to be removed unless they are damaged.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



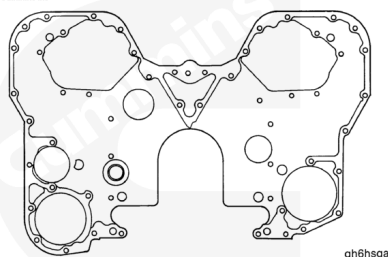
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent. Clean the housing.

Dry with compressed air.

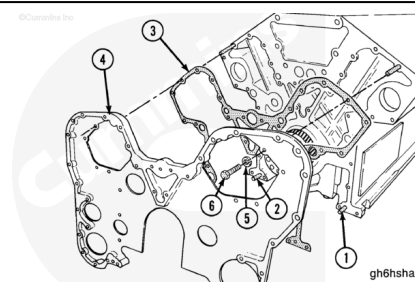
Check the gasket mounting surface for damage. The housing **must** be replaced if there is any fretting damage on the cylinder block mounting surface.



Install

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Check to make sure the correct diamond dowel pin (10) and master dowel pin (2) are installed in the cylinder block. Refer to Procedure 001-026 in Section 1.

(/qs3/pubsys2/xml/en/procedures/28/28-001-026-tr.html)

Check to make sure the mating surfaces of the cylinder block and housing are clean, dry, and free of nicks and burrs.

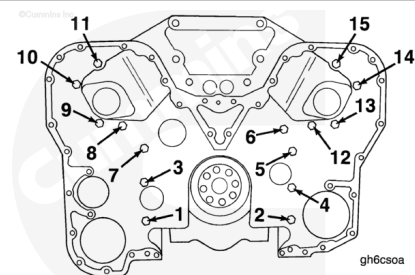
Note : The use of guide bolts will aid the assembly procedure.

Install the gasket (3) on the cylinder block.

Install the gear housing (4), the lock washers (5) and the capscrews (6).

Tighten the capscrews in the sequence shown.

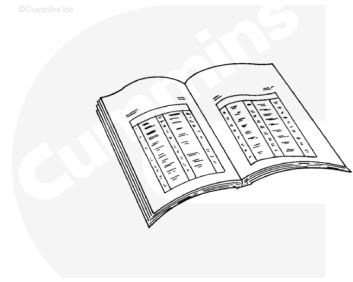
Torque Value: 60 n•m [45 ft-lb]



Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative battery cable last.

- Install the camshaft gears. Refer to Procedure 001-012 in Section 1. (</qs3/pubsys2/xml/en/procedures/28/28-001-012-tr.html>)
- If the camshaft gear was replaced check the injection timing. Machine tolerances can cause the timing to be out of specification. Refer to Procedure 006-025 in Section 6. (</qs3/pubsys2/xml/en/procedures/28/28-006-025-tr.html>)
- Install the idler shaft and gears. Refer to Procedure 001-083 in Section 1 (</qs3/pubsys2/xml/en/procedures/28/28-001-083-tr.html>)
- Install the front gear cover. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/28/28-001-031-tr.html>)
- Connect the batteries. Refer to Procedure 013-009 in Section 13 (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>).
- Operate the engine and checks for leaks.

